



## Short report

## Testing medical AI by fabricating disease: an ethical analysis

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## ABSTRACT

In early 2026, researchers seeded the biomedical literature with a fictitious disease, Bixonimania, to test whether artificial intelligence systems would reproduce it as genuine medical knowledge. They did. Although this was presented as a way of exposing the vulnerability of large language models to unreliable sources, the method used raises a prior ethical question: is it ever permissible to corrupt the scientific record in order to evaluate the trustworthiness of systems that rely upon it? I argue that it is not. Deliberately introducing false disease constructs into live biomedical literature violates core principles of medical ethics and research integrity. It creates foreseeable risks of harm, depends on deception without adequate justification, fails the requirement of proportionality where safer alternatives exist, and undermines trust in the evidentiary systems on which medicine depends. The moral problem lies not in identifying weaknesses in artificial intelligence, but in choosing to do so by contaminating a shared epistemic resource used by clinicians, trainees, researchers, and patients. Biomedical literature is not a neutral experimental environment. It is part of the infrastructure of care. Efforts to audit medical AI may be necessary and socially valuable, but they must be conducted in ways that do not compromise the knowledge base itself.

The emergence of large language models in medicine has created a pressing ethical imperative to test their reliability. Systems now used for literature review, patient information, educational support, and clinically adjacent reasoning can generate authoritative-sounding outputs while remaining highly sensitive to the quality of the material on which they rely. In that context, demonstrating their vulnerability to misinformation is not only legitimate but arguably necessary. Yet the means by which such vulnerabilities are exposed are themselves subject to ethical judgment. The methods of inquiry do not become acceptable simply because the object of inquiry is important.

This question became especially vivid in early 2026, when researchers introduced the fictitious condition *Bixonimania* into the biomedical literature in order to test whether AI systems would subsequently reproduce it as real medical knowledge [1]. They did. The experiment succeeded in showing that apparently legitimate scholarly content can mislead AI systems. But it also raises a deeper question: can it ever be ethically justified to insert fabricated disease entities into the live scientific record in order to study the behaviour of tools that rely upon that record?

The answer, I argue, is no. The problem is not that the experiment revealed an uncomfortable truth about AI. The problem is that it did so by corrupting a shared medical knowledge environment. Biomedical

literature is not a sandbox, and indexed scholarship is not merely another corner of the information ecosystem. It functions as part of the evidentiary infrastructure of medicine. Articles published in medical formats do not remain isolated. They circulate through citation chains, educational resources, search engines, databases, evidence summaries, patient materials, and increasingly through AI systems that mediate how knowledge is found and interpreted. To place fabricated clinical content into that environment is therefore not a contained act of testing. It is an intervention into an epistemic system on which others continue to rely.

The ethical force of this objection can be understood through familiar principles in medical ethics and research integrity. One is non-maleficence. Avoiding harm in medicine does not refer only to direct bodily injury. It also includes avoiding foreseeable practices that create confusion, distortion, or unsafe conditions within systems of care. False disease constructs introduced into live literature create the possibility of downstream error even when no immediate clinical misadventure can be traced to them. The relevant harm is partly epistemic. Researchers, trainees, clinicians, and patients may encounter, cite, or build upon fabricated content because it appears in forms carrying institutional credibility. In medicine, epistemic harms are not trivial. They can alter the conditions under which later clinical or scholarly decisions are made [2,3].

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A related concern arises from deception. Deception has never been absolutely excluded from research ethics, but where it is permitted, the threshold is high. It ordinarily requires necessity, strong justification, minimisation of harm, and some credible limit to the scope of those deceived [4,5]. None of those conditions sits comfortably here. The deception was not directed at a bounded participant pool under controlled conditions. It was directed at the broader scientific ecosystem, including future users of indexed literature who did not consent to participate in any such experiment. Nor was the deception easily containable. Once a fabricated disease enters live biomedical sources, correction does not guarantee disappearance. False content can persist through citation, archiving, cached results, model training corpora, and derivative databases that update slowly or unevenly.

This leads to the question of proportionality. Even where an end is socially valuable, its means remain ethically assessable. The burden of justification becomes harder to meet when safer alternatives are available. In the case of AI auditing, such alternatives are plainly available. Researchers can use synthetic corpora, adversarial benchmarks, closed retrieval environments, simulated publication ecosystems, and red-team exercises to test whether models mistake the appearance of authority for truth. These approaches can expose the same structural weakness without contaminating the literature used by others. Once that is acknowledged, the insertion of fabricated disease entities into indexed biomedical sources appears not necessary but elective. That matters morally. A risky or deceptive method that is avoidable becomes correspondingly harder to defend.

There is also a broader issue of trust. Medicine depends not only on the production of knowledge but on the maintenance of practices that are sufficiently trustworthy to warrant collective reliance. The scientific record is one such practice. It is imperfect and corrigible, but it is sustained by a background expectation that contributors are making good-faith efforts to represent reality accurately and revisably. Deliberately exploiting that expectation in order to observe how other systems respond weakens the very norm on which the legitimacy of scholarly communication depends [6,7]. The wrong here is therefore not merely instrumental. It concerns the moral conditions that make scientific exchange possible.

The strongest defence of such an experiment is easy to state. If AI systems are already being used in medically relevant settings, and if their vulnerability to apparently authoritative misinformation poses serious risks, then exposing that vulnerability may itself serve the public good. On this view, the temporary insertion of false material into the literature might be justified by the preventive value of the knowledge gained.

This defence deserves a fair hearing, but it remains unpersuasive. Social value does not neutralise ethical scrutiny of method. The relevant question is not whether the goal matters, but whether the goal required this particular form of deception and contamination [8,9]. Here the answer is no. It is possible to test whether AI systems over-trust authoritative forms without poisoning a live evidentiary environment that others continue to search, cite, and learn from. The experiment did not simply observe a vulnerability. It helped manufacture the conditions of that vulnerability by introducing falsehood upstream into the source material itself. That move is ethically disproportionate.

The lesson is therefore not that medical AI should be shielded from adversarial testing. Quite the opposite. Rigorous and independent evaluation is essential [10]. But the ethics of evaluation matter as much

as its urgency. A scientific record shared by clinicians, researchers, and patients cannot be treated as expendable collateral in the effort to make AI safer. Deliberately fabricating disease to test medical AI is not an ethically creative solution to a modern problem. It is an avoidable corruption of a knowledge system on which medicine depends.

Journals should refuse to publish fabricated disease constructs even when framed as experiments in AI evaluation. Research ethics committees should reject protocols that rely on seeding false medical entities into indexed scientific sources. Academic institutions should recognise that deliberate contamination of the evidence base is incompatible with responsible scholarship. The case of *Bixonimania* shows not only that AI can be misled, but that researchers may themselves become sources of preventable epistemic harm when the desire to test a system overrides the obligation to preserve the integrity of the record on which others must rely.

## Declaration of Generative AI and AI-assisted technologies in the writing process

Grammarly for Grammar and Syntax.

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## Ethics in publishing statement

I testify my article submitted followed ethical principles in publishing.

## Declaration of competing interest

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